

LM: Term Structure & Interest Rate Dynamics

Benchmark Curve 给目标收益率选 Benchmark.

$$y = y_B + \text{spread}$$

1. Spot Curve.

① S & YTM 关系:

Coupon-paying	zero-coupon
$P = \frac{CF_1}{1+s_1} + \frac{CF_2}{(1+s_2)^2}$	$P = \frac{CF_T}{(1+s_T)^T}$
$= \frac{CF_1}{1+YTM} + \frac{CF_2}{(1+YTM)^2}$	$= \frac{CF_T}{(1+YTM)^T}$
YTM 相当于一系列 S 平均, 更接近到期日 S	$S = YTM$

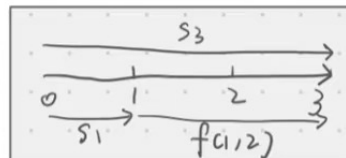
2. Forward's Curve

$f(1,2)$ 代表 1-3.
 ↑ ↑
 开始时期 期限

① 折现因子 Discount Factor: $DF(1,1) = \frac{1}{1+f(1,1)}$ $DF(1,2) = \frac{1}{[1+f(1,2)]^2}$

② Forward Rate Model 可视为面值为1的 Forward Price.

① 计算 $(1+s_3)^3 = (1+s_1) \cdot (1+f(1,2))^2$
 $S \iff f$ 要能够相互求解.



② 结论:
 { spot curve $\nearrow$, forward curve 也 $\nearrow$, 且在 spot curve 之上 $\implies f \rightarrow$ 满足 riding yield down
 { spot curve $\searrow$, forward curve 也 $\searrow$, 且在 spot curve 之下 $\implies f$

③ Forward Pricing Model 远期合约定价 (求 Forward Contract Price, 本折现因子)

利用 Arbitrage Free Pricing: $DF_B = DFA \times F_{A,B-A}$

$$F_{A,B-A} = \frac{DF_B}{DFA}$$

F, DF 代表 0 时点价格, 也是折现因子.

④ Active Bond Portfolio Management

① 预期 spot rate 与 forward rate 对比 $\left\{ \begin{array}{l} s' = f \\ s' < f \rightarrow P_B \uparrow \text{ 标的价格上升} \rightarrow \text{long} \\ s' > f \rightarrow P_B \downarrow \text{ 标的价格下降} \rightarrow \text{short} \end{array} \right.$

② Riding Yield Curve.
 { 假设: yield curve $\nearrow$ 且 稳定. maturity $\uparrow$. 都导致超额收益 $\uparrow$
 { 结论: buy bonds with a maturity longer than investment horizon.
 curve steeper

3. Par Curve.

① Par Rate 定义: Bond Price 平价发行, Price = Par Value 的 Par Bond, $YTM = \text{Coupon Rate} = \text{Par Rate}$

② 计算: $PR_t \iff S_t$ Bootstrapping 法. 从前往后算.

已知 PR 求 S	
1yr: $P = \text{Par} = \frac{C_1 + \text{Par}}{1+s_1}$	2yr: $P = \text{Par} = \frac{C_2}{1+s_1} + \frac{C_2 + \text{Par}}{(1+s_2)^2}$
$C_1 = \text{Par} \times CR_1 = \text{Par} \times PR_1$	$C_2 = \text{Par} \times CR_2 = \text{Par} \times PR_2$
$s_1 = PR_1$	代入反求 s_2 .

4. YTM (非重点)

① YTM & S 的关系

zero coupon : $S = YTM$
 coupon paying : YTM 相当于一系列以 CF 为权重的 S 的加权平均。

★ 5. Swap Rate Curve

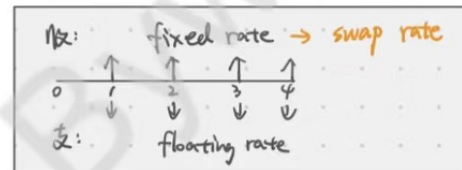
① 计算

$$V_{\text{swap}} = V_{\text{fixed}} - V_{\text{floating}} = 0$$

$$V_{\text{fixed}} = V_{\text{floating}} = \text{Par Value. 平价互换.}$$

$$\text{Par} = \frac{C}{1+S_1} + \frac{C}{(1+S_2)^2} + \frac{C}{(1+S_3)^3} + \frac{C+\text{Par}}{(1+S_4)^4} \Rightarrow \frac{C}{\text{par}} = \text{Coupon Rate} = \text{Swap Rate}$$

Swap Rate 本质是 Par Rate, 都 = coupon Rate.



② 估值

Liquidity.

相较于国债 期限更多选择

Swap 大型金融机构较多选择, 作为目标公司债的 benchmark curve 相似度更高。

OTC 交易 + 可跨国, unregulated.

V.S. Government Spot Curve

③ V.S. 国债

Wholesale Bank, 多以 swap curve 作 benchmark

Detail Bank 面对散户, 多以 government spot curve 作 benchmark.

Yield Spread.

$$i = y_B + \text{spread} \rightarrow \text{spread} = y_i - y_B$$

1. Swap Spread.

① Swap Spread 以 Swap 作为标的, 以 on the run 国债为 benchmark

$$\text{swap spread} = \text{swap rate} - \text{treasury yield} \quad (\text{same maturity}). \quad \text{大型企业相较于国企}$$

② 风险补偿

high default risk

market liquidity (美国国债流动性高).

2. I-Spread.

① I-Spread = Bond Rate - Swap Rate. (same maturity)

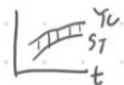
标的公司相较于大型企业

可理解为线性差补方法. $y_{\text{长}} = y_{\text{短}} + \text{Spread} \cdot t$

② 风险补偿: credit and liquidity risk.

3. Z-Spread 更反应公司债风险情况.

视为常数!!



① 公式: $Z\text{-spread} = S_{\text{company}} - S_{\text{treasury}} \quad (\text{same maturity}).$

② 风险补偿: credit, liquidity and option risk (可转债含权). 但 $Z\text{-spread} = \text{Zero volatility}$, option 不准确.
 对公司债风险补偿衡量: $Z\text{spread}$ 比 swap spread 更准确.

4. TED Spread 更反应银行风险. (信用风险).

① 公式: $\text{TED} = \text{LIBOR} - \text{T-Bill Rate.} \quad (\text{same maturity}).$

② 风险补偿. T-bill: 政府 $\Rightarrow$ credit and liquidity risk
 LIBOR: 银行 $\Rightarrow$ + default risk in banking sector.

5. LIBOR-OIS Spread 更反应短期市场风险 (流动性).

① 公式: $\text{Spread} = \text{LIBOR} - \text{Overnight Indexed Swap (OIS) Rate (无抵押)}$
 usually 90 day 1-day

(different maturity).

② 风险补偿: Money Market Security Risk. (maturity 导致).

4 Term Structure theories.

解释模型曲线

1. Pure (Unbiased) Expectation Theory 纯预期理论

- ① 描述
- ① 长期利率是短期利率几何平均. Forward Rate 是未来短期利率的无偏预期: $f = E(S')$
 - ② Unbiased Expectation 成立, 意味投资者 Risk-Neutral
- ② Shape: $R_2 = \frac{r_1 + r_2^e}{2}$ (简化), 理应是几何平均: $R_2 = \sqrt{(1+r_1) \cdot (1+r_2^e)} - 1$.

risk neutral x

$$\begin{cases} \textcircled{1} \text{ If } r_2^e \uparrow \rightarrow R_2 \uparrow > r_1 \rightarrow \nearrow \\ \textcircled{2} \text{ If } r_2^e \downarrow \rightarrow R_2 \downarrow < r_1 \rightarrow \searrow \\ \textcircled{3} \text{ If } r_2^e = \rightarrow R_2 = r_1 \rightarrow \rightarrow \end{cases}$$

Local Expectation Theory < short term risk-free.
局部预期理论 long term $\checkmark$ risk.

2. Liquidity Preference Theory

- ① 描述: 预期有偏, + 流动性风险补偿. $f = E(S') + \text{Liquidity Premium (LRP)}$

② Shape: $R_2 = \frac{r_1 + r_2^e}{2} + \text{LRP}$

$$\begin{cases} \textcircled{1} \text{ If } r_2^e \uparrow \rightarrow R_2 \uparrow > r_1 \rightarrow \nearrow \\ \textcircled{2} \text{ If } r_2^e = \rightarrow R_2 = r_1 \rightarrow \rightarrow \\ \textcircled{3} \text{ If } r_2^e \downarrow \left\{ \begin{array}{l} \downarrow \text{幅度小} \rightarrow R_2 > r_1 \rightarrow \nearrow \\ \downarrow \text{幅度大} \rightarrow R_2 = r_1 \rightarrow \rightarrow \\ \downarrow \text{幅度巨大} \rightarrow R_2 < r_1 \rightarrow \searrow \end{array} \right. \end{cases}$$

解释 $\rightarrow$ 最好的理论
但很难解释 $\rightarrow$

3. Segmented Market Theory

- ① 描述
- ① 利率由资金市场 supply and Demand 决定 $\Rightarrow \begin{cases} r_{\text{短}}: D_s \& S_s \\ r_{\text{长}}: D_L \& S_L \end{cases}$
 - ② 不同期限的资金市场完全分割.

② Shape

- ① $\nearrow: r_s < r_L \rightarrow D_s < S_s; D_L > S_L$
- ② $\searrow: r_s > r_L \rightarrow D_s > S_s; D_L < S_L$

对比

市场分割: horizon 不变.
偏好: horizon 改变.

4. Preferred Habitat Theory

- ① 描述: 擅长投短期, $r_s \uparrow \rightarrow$ 会其投长期, + premium. r_L 大

② Shape

- ① $\nearrow: \text{prefer invest short} \xrightarrow{+r \uparrow} \text{invest long}, r_s < r_L$
- ② $\searrow: \text{prefer invest long} \xrightarrow{+r \uparrow} \text{invest short}, r_s > r_L$

Yield Curve Factor Models = Yield Curve Risk 形状改变的风险.

- Yield curve 形状变化影响因素.
 - ① Level, 只要同上同下变动都算. 最常见. Δx_L
 - ② Steepness, non-parallel, 长短期变化幅度不同. 只提及长、短期 Δx_S
 - ③ Curvature, 长、中、短期都变动 Δx_C

2. Managing Yield Curve Risk.

① Effective Duration: parallel 各时段 Δy 相同

② Key Rate Duration: non-parallel

$$KRD_i = \frac{-\Delta P/P}{\Delta y_i}$$

- 定义: 将 curve 理解为不同 maturity 的组合. KRD measure sensitivity to a change in specific maturity segment.
- 公式:
 - $KRD_i = W_i \times D_i$
 - $D_p = \sum KRD_i$ 影响如何高.
- 计算: 非平行移动 $\rightarrow \Delta P/P$ (分割思想)

y 增加时, value 减小, 反向

$$\Delta P/P = -[KRD_1 \times \Delta y_1 + KRD_2 \times \Delta y_2 + \dots]$$

- 按期限分 $D_p = -\frac{\Delta P}{\Delta y} \rightarrow \Delta P/P = -D_p \times \Delta y$
- 按影响因素分: $\frac{\Delta P}{P} = -D_L \cdot \Delta x_L - D_S \cdot \Delta x_S - D_C \cdot \Delta x_C$
Level + steepness + curvature

3. Yield Curve Volatility.

① Reasons.

- Embedded Option
- Risk Management 的内容.

② Short-Term rates $\downarrow$ > Long-Term rates $\downarrow$

③ 平方根法则: $\sigma_{\text{month}} = \sigma_{\text{annual}} \cdot \sqrt{\frac{1}{12}}$



Macroeconomic Variable

$$\text{Bond Risk Premium (Term Premium)} = y_L - y_{\text{短}} \quad (\text{国债})$$

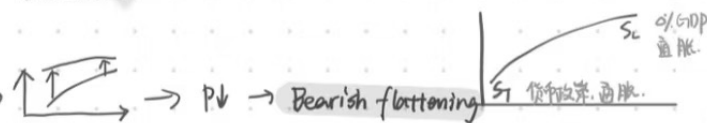
benchmark

经济好, spread $\uparrow$, 陡峭
经济差, spread $\downarrow$, 倒挂

1. 宏观影响 Yield curve 因素

① Monetary Policy

经济扩张: 紧缩. 对经济不好. $r_s \uparrow \rightarrow r_L \uparrow \rightarrow P \downarrow \rightarrow \text{Bearish flattening}$
经济萧条: 扩张. 对经济好. $r_s \downarrow \rightarrow r_L \downarrow \rightarrow P \uparrow \rightarrow \text{Bullish steepening}$



② Fiscal Policy

财政赤字 $\uparrow$ ($G > T$) $\rightarrow D \uparrow \rightarrow$ 发债 $S_B \uparrow \rightarrow P_B \downarrow \rightarrow y_B \uparrow$; 反之 $y_B \downarrow$.

③ Maturity Structure.

Segmented Market Factor. $S_L \uparrow \rightarrow P_L \downarrow \rightarrow y_L \uparrow$. 长期、短期互不影响. 只受者 D & S 影响.

④ Investor Demand.

① Domestic Demand

$\rightarrow \text{Demand} \uparrow \rightarrow y \downarrow$

② Non-Domestic Inflow.

③ 经济不好时, flight $\leftarrow$ quality. (放弃风险资产 转而买国债) $\rightarrow P_L \uparrow \rightarrow y_L \downarrow \rightarrow \text{bullish flattening}$



2. Interest Rate Management.

调整 Duration: $r \uparrow$, 为了让 P 下降, 应降低 Duration. (Sell 长 Bond Buy 短 Bond)

$r \downarrow$, $P \uparrow$, 应增加 Duration.

3. 交易策略.

① Duration Neutral: $\Delta D = 0$.

When Steeper Curve: 买短债, 卖长债.
When Curve Flatter: 买长债, 卖短债

$r \uparrow$, $P \downarrow$: short.
 $r \downarrow$, $P \uparrow$: long.

② Long Only: Bullet.
Barbell



适用于预期 Bullish Flattening



以获得两端收益.

2M2. Arbitrage-Free Valuation

Binomial Interest Rate Tree.

1.  $i_H = i_L e^{2\sigma}$ 同一支, 高低相差2倍 volatility.
2. Volatility 变动 $\left\{ \begin{array}{l} \sigma \uparrow, \text{ spread out 高低差变大} \\ \sigma \downarrow, \text{ 收敛, if } \sigma \rightarrow 0, i_H = i_L \end{array} \right.$

3. 构建 r 二叉树.

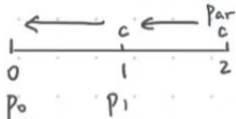
$$r \sim \log N \\ \sigma \text{ constant}$$

Middle Rate.

- ① 一期: 求 forward rate, 作为 i_m $\left\{ \begin{array}{l} i_H = i_m e^{\sigma} \\ i_L = i_m e^{-\sigma} \end{array} \right.$
- ② 二期: 求 forward rate, 作为 i_m $\left\{ \begin{array}{l} i_H = i_m e^{2\sigma} \\ i_L = i_m e^{-2\sigma} \end{array} \right.$



Option-Free Bond Valuation



从末期往前推, 每期折现率 r_H, r_L 各 50% 概率.

Postwise Valuation. 路径估值. 各路径单独计算取平均.

Monte Carlo Method. $\rightarrow$ 用于 MBS 估值

1. MBS 特征 $\left\{ \begin{array}{l} ① \text{ 有 CF 路径依赖特征.} \\ ② \text{ 不能用 } r \text{ 二叉树估值. 用 Monte Carlo.} \end{array} \right.$
2. Monte Carlo Method. $\left\{ \begin{array}{l} \text{N个路径 } n \text{ 个 CF, CF 折现求平均} \end{array} \right.$

Term Structure Model $\rightarrow$ 路径模拟模型.

1. Equilibrium Term Structure

- ① One Factor Model (r , single factor)
- ② Multiple Factors Model (r, r_m, r_c).

① Cox-Ingersoll-Ross Model: $dr_t = k(\theta - r_t)dt + \sigma\sqrt{r_t}dz$

结论: $\left\{ \begin{array}{l} 2 \text{ 部分组成: 趋势 } dt \text{ or } \theta t. \text{ 波动 } dz \text{ or } \Delta z \end{array} \right.$

θ 代表 r_c , mean-reversion 特征.

波动项系数 $\sigma\sqrt{r_t}$ 代表波动项受 r 影响.

② Vasicek Model: $dr_t = k(\theta - r_t)dt + \sigma dz$

结论: 与 CIR 比, 波动项系数不同 不受 r 影响.

① Ho-Lee Model: $dr_t = \theta_t dt + \sigma dz$

② KWF Model: $d\ln(r_t) = \theta_t dt + \sigma dz$

相同点: θ_t 是常数, no mean-reverting, volatility 常数.

不同点: KWF 用 $\ln(r_t)$ 服从正态分布, r_t 服从对数正态分布.

Arbitrage Opportunity (区分)

Type: $\left\{ \begin{array}{l} \text{Value Additivity } 4P_A \approx 1P_B. \text{ 存在倍系数} \\ \text{Dominance. } r_A \approx r_B. \text{ 根据 } P \text{ 买低卖高.} \end{array} \right.$

Estimation $\left\{ \begin{array}{l} \text{Historical} \\ \text{Observed.} \end{array} \right.$

Exhibit 33 Term Structure Model Summary

Model	Type	Short Rate	Drift Term	Volatility
CIR	Equilibrium	dr_t	Mean reversion at speed k	Varies with $\sqrt{r_t}$
Vasicek	Equilibrium	dr_t	Mean reversion at speed k	Constant
Ho-Lee	Arbitrage free	dr_t	Time dependent	Constant
KWF	Arbitrage free	$d\ln(r_t)$	Time dependent	Constant

Summary.

基础.

伊藤定律

$\Delta r = \text{趋势项} + \text{波动项}$

$\Delta r = m \cdot \Delta t + n \cdot \Delta z$

随机, 假设服从某分布

长 趋势项 波动项

$r_c = 1$

类似于 $d\ln(r_t)$

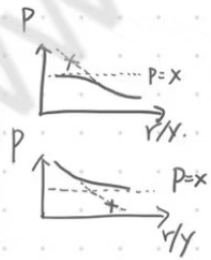
波动项相同.

HL 有 θ_t 市场参数, closely fit observed mkt yield.

LM3. Bonds with Embedded Option

含权债券回顾.

1. Callable Bond: Issuer 提前赎回 $\rightarrow P \downarrow Y \uparrow$ $V_{callable} = V_{pure} - V_{call}$
2. Puttable Bond: Bondholder 提前赎回 $\rightarrow P \uparrow Y \downarrow$ $V_{puttable} = V_{pure} + V_{put}$
3. Convertible Bond: Bondholder 债转股 $\rightarrow S > X$, 类似于 Callable Bond $V_{convertible} = V_{pure} + V_{call \text{ on stock}}$
4. Bonds with Estate Puts: Bondholder 债券人以 Par Value 卖回 Issuer. 寿命越短, V_{put} 越大.
5. Sinking Fund Bond 偿债基金: credit risk.



Valuation of Callable, Puttable Bond.

1. 二叉树估值 } Callable Bond, $P > X$ 时, 取 X
Puttable Bond, $P < X$ 时, 取 X 从末期倒推求解.
2. Embedded Option 估值 (仍以二叉树为方法). } $V_{call} = V_{pure} - V_{callable}$
 $V_{put} = V_{puttable} - V_{pure}$

① 二叉树求出 Embedded Option Bond Value. 同上.

② 再由公式 $V_{Bond} - V_{pure}$.

3. Effect of Interest Rate Volatility.

Call 反向. Put 正向.

常考 $\star$ $\Delta r \rightarrow$ Embedded Option $\rightarrow$ Bond Value 不同. $V_{callable} \downarrow \rightarrow OAS \text{ call} \downarrow \rightarrow \text{option cost} \uparrow$
 $V_{puttable} \uparrow \rightarrow OAS \text{ put} \uparrow \rightarrow \text{option cost} \uparrow$

4. OAS 计算. 常数, 加在每个节点使理论 = 参考.

① 辨析 } OAS: 剔除权利后 (行权后) spread, 受 Δr 影响.
ZS: 持有到期 (未行权) 的 spread.

Callable Bond: $OAS < ZS$ $V_{call} = ZS - OAS$

Puttable Bond: $OAS > ZS$ $V_{put} = OAS - ZS$

② 计算: $P_{理论} \neq P_{MKT}$.

试错法: 二叉树每个节点 + OAS. 考试先试 Middle OAS.

③ 结论 ① 含权债券风险补偿. OAS 最准确.

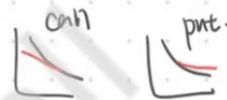
② 相同特征 Bond, OAS 应相似 } $OAS \uparrow \rightarrow r \uparrow \rightarrow P \downarrow$ underprice.
 $OAS \uparrow \rightarrow r \downarrow \rightarrow P \uparrow$ overprice

relative valuation

④ $\Delta r \rightarrow OAS$: Δr 不变 ZS 不受 Δr 影响 } $V_{call \text{ option}} \uparrow = ZS - OAS \rightarrow OAS \downarrow$
 $V_{put \text{ option}} \uparrow = OAS - ZS \rightarrow OAS \uparrow$

5. IR 变动的影响.

① } callable: $r \uparrow \rightarrow P \downarrow$
puttable: $r \downarrow \rightarrow P \downarrow$
call option } $r \uparrow$ 不行权 $V_{call} \downarrow$ OTM
 $r \downarrow$ 行权 $V_{call} \uparrow$ ITM
put option } $r \uparrow$ 行权 $V_{put} \uparrow$ ITM
 $r \downarrow$ 不行权 $V_{put} \downarrow$ OTM



$r \downarrow$ 对应 upward yield curve flatten

$\hookleftarrow$ 平坦 $\rightarrow$ Average $r \downarrow$ 边缘 $r \downarrow$

$V_{call} \uparrow$
 $V_{put} \downarrow$

② Effective Duration & Convexity

① 计算 $\left\{ \begin{array}{l} \text{计算 OAS, 代入 OAS 算新二叉树, 得 } V_0 \\ \text{每节点 } -0.5\%, \text{ 算 } V_- \\ \text{每节点 } +0.5\%, \text{ 算 } V_+ \end{array} \right\}$ 上下变动量相同

② 结论 $\left\{ \begin{array}{l} \text{代入 ED 或 EC 公式计算} \\ \text{ED 可近似于 MACD (平均还款期)} \\ \text{ED 4 种情况} \\ \text{EC } \left\{ \begin{array}{l} \text{Straight Bond: 凹性, 涨多跌少} \\ \text{Callable Bond: } \sim \\ \text{Puttable Bond: } \sim \end{array} \right. \end{array} \right.$

$$ED = \frac{(V_-) - (V_+)}{V_0} / 2\Delta y$$

$$EC = \frac{(V_-) + (V_+) - 2 \times (V_0)}{\Delta y^2 \cdot V_0}$$

③ One-Side Duration (up or down)

Callable Bond:
 $r \downarrow$ P 升 $r \uparrow$ P 降
 Down Duration 小 < Up Duration 大

Puttable Bond
 $r \downarrow$ P 升 $r \uparrow$ P 降
 Down Duration 大 > Up Duration 小

④ Key Rate Duration (specific maturities)

Identify shaping Risk for Bond.

凸性质: 不含权

- For option-free bond that is traded at par
 - Bond's maturity matched rate is the only rate that affects the bond's value
 - All other key rate durations are zero
- For option-free bond that is not traded at par
 - Shift the maturity-matched par rate has the greatest effect
- For zero-coupon bond or bond with a very low coupon
 - Key rate durations can sometimes be negative for maturity points that are shorter than the maturity of the bond being analyzed

Traded at par. $S \uparrow$, S 可能 $\downarrow$

含权:

3. For callable bonds

① With low coupon rate

- The callable bonds are unlikely to be called
- The rate that has the highest effect on the value of the callable bond is the maturity-matched rate

② As the bond's coupon increase

- So does the likelihood of the bond being called
- The rate that has the highest effect on the callable bond's value is the time-to-exercise rate

4. For puttable bonds

① With high coupon rate

- The puttable bonds are unlikely to be put
- Prices are most sensitive to their maturity-matched rates

② As the bond's coupon decrease

- So does the likelihood of the bond being put
- price is most sensitive to the time-to-exercise rate

Capped or Floored Floating Rate Bond (Floater).

1. Definition and Application

pay in arrears & CR = LIBOR 时. option-free bond 在利息重置日 Par Value = Bond Price.

Cap: Coupon Rate 上限, 利于 Issuer. r 越高 Cap 越有价值 $\approx$ pure floater + Call Option (看涨 r) Investor 卖给 Issuer
 Floor: Coupon Rate 下限, 利于 Investor. r 越低 Floor 越有价值 $\approx$ pure floater + Put Option (看跌 r) Issuer 卖给 Investor

Floater with cap

A $\longrightarrow$ B
 sell straight bond buy s.b.
 buy call option sell call option.

$$V_{\text{capped floor}} = V_{\text{s.b.}} - V_{\text{call option on } i}$$

$$V_{\text{callable bond}} = V_{\text{s.b.}} - V_{\text{call option 赎回权}}$$

Floater with floor

A $\longrightarrow$ B
 sell s.b. buy s.b.
 sell put option buy put option

$$V_{\text{floor floater}} = V_{\text{s.b.}} + V_{\text{put option on } i}$$

$$V_{\text{puttable bond}} = V_{\text{s.b.}} + V_{\text{put option 售出权}}$$

2. 二叉树 Calculation 先比较再计算.

$$\text{Value of Cap} = \text{Value of Straight Floater} - \text{Value of Capped Floater}$$

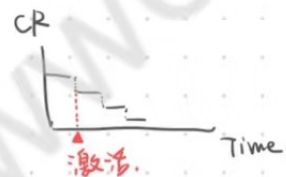
$$\text{Value of Floor} = \text{Value of Floored Floater} - \text{Value of Straight Floater}$$

满足条件时 $\left\{ \begin{array}{l} \text{Pay in arrears} \\ \text{CR} = \text{LIBOR} \\ V = \text{Par value} \end{array} \right.$

3. Ratchet Bond 棘轮债券

Floating Rate Bond with both issuer and investor option.

- Issuer: Cap 浮动, 保持不变或下调. 跟随 Coupon Rate 变动.
- Investor: 只要调整, investor 的 put option (或有回售权) 被激活.



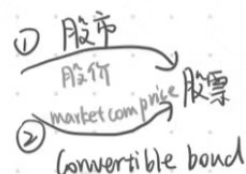
Convertible Bond

1. Basic Concepts.

债转股, a specified number conversion ratio. at a pre-determined conversion price

- Conversion price = Issue price / conversion ratio.
conversion ratio = Issue price / conversion price

- Market conversion price = Convertible bond price / Conversion ratio
在今天通过可转债的方法得到股票的成本.
Market conversion premium per share = Market conversion price - Market share price = ② - ①.
Market conversion premium ratio = Market conversion premium per share / Market share price



- Conversion Value = Market share price x Conversion ratio. 股票当前总市值.
- Straight Value = PV of Cash Flows 对可转换权利视而不见
- Minimum Value = Max (Conversion Value, Straight Value). 保底价值, 两个极端取其高.
当下转 并不转.

2. Analysis

① 风险收益特征

① 上不封顶, 下有保底

- Upside potential: 体现股性
- Downside Risk: 体现债性. 等于 Straight Value. Downside Risk $\in$ Minimum Value, 只考虑稳健持有.

$$\text{Premium over straight value} = (\text{Convertible bond price} / \text{straight price}) - 1$$

② 与直接买股票相比

a. Stock price increases

- Convertible bond will underperform
- Reason: conversion premium

b. Stock price decreases

- Returns on convertible bonds exceed those of the stock
- Reason: The convertible bond's price has a floor value

c. Stock price remains stable

- Returns on convertible bonds exceed those of the stock
- Reason: The coupon payments received from the bond, assuming no change in interest rates or credit risk of the issuer

- Underlying share price well below conversion price. $\rightarrow$ 考虑“债”性
Underlying share price well above conversion price. $\rightarrow$ 考虑“股”性
between 2 extremes, hybrid.

② Valuation Equivalence

$$\text{Convertible Bond} = \text{Straight Bond} + \text{Call Option on Stock} / (\text{Put option on Bond})$$

$$V_{\text{convertible bond}} = V_{\text{sb}} + V_{\text{转换权}}$$

LM4 Credit Analysis Models.

Modeling Credit Risk

1. 基本名词

Expected Exposure 风险敞口 $Exposure_t = P_t + C_t$

Probability of Default.

Recovery Rate.

Loss Given Default $1 - R = 1$.

Expected Loss $EL = (1 - POD) \times 0 + POD \times LGD = POD \times LGD$.

Credit Valuation Adjustment $CVA = PV \text{ credit value} = \sum PV EL$

Value No Default (VND): Fair Value of Bond = VND - CVA.

	EL_1	EL_2	EL_3
	$POD_1 \times LGD_1$	$POD_2 \times LGD_2$	$POD_3 \times LGD_3$
	1	2	3
$PV = PV EL_1 + PV EL_2 + PV EL_3$			

2. POD

Actual POD

Risk-Neutral POD: 风险中性定价概率求解的 POD.
分子考虑风险, 分母 Hrf 折现.

通常 risk neutral POD > actual pod.

Valuation Credit Spread

1. 四种情况

同-个 r = 二叉树	Zero-coupon Bond
	Coupon-paying Bond
	Fixed-rate Bond
	Floating-rate Bond

→ 考试可放弃.

		$\text{Exposure} = P_t + C_t$	
同-个 r	{	Zero-coupon	$P_t \quad 0$
		coupon paying	$P_t \quad C_t$
		fixed rate	$\bar{P}_t \quad C_t$
		floating rate	$\bar{P}_t \quad \bar{C}_t$
= 叉树			

$E_t = \text{利率折现} + \text{当期 CF}$

2. 过程

① $FV = VND - CVA$ $\left\{ \begin{array}{l} VND \text{ 未违约, 直接估值. } = \sum PV CF_t \\ CVA = \sum PV EL_t = \sum EL_t \cdot DF_t \end{array} \right.$

$EL_t = POD_t \times LGD_t$

- $LGD_t = Exposure_t \times (1 - RR)$
 $Exposure_t = P_t + C_t$ $< P_t$ 为利率折现求和
- $POD_1 = POD$, $POS_1 = 1 - POD$
 $POD_2 = POS_1 \times POD$, $POS_2 = 1 - POD_2$
二叉树 P_t 为节点加权平均
Floating P_t, C_t 都加权平均

② Rate of Return 到期: YTM, 由 FV 求.

违约: $IRR = \sum \text{每期收益} / (1 + IRR)^t = \text{期初投资}$
违约: 收益 = Recovery
不违约: 收益 = 0 or coupon

③ Credit Spread = $YTM_c - YD$

3. 二叉树 Volatility

$r \sim \log N$. 分布不对称. $6\uparrow \rightarrow r\uparrow, r\downarrow$.

$6\uparrow$ $\left\{ \begin{array}{l} r\uparrow \\ r\downarrow \end{array} \right.$ $\rightarrow P\downarrow \rightarrow Exposure\downarrow \rightarrow LGD\downarrow \rightarrow EL\downarrow \rightarrow CVA\downarrow$
 $6\downarrow$ $\rightarrow P\uparrow \rightarrow Exposure\uparrow \rightarrow LGD\uparrow \rightarrow EL\uparrow \rightarrow CVA\uparrow$
 $CVA\downarrow \rightarrow FV\uparrow$

Models.

1. Credit Score

FICO 费埃哲 Score.

2. Credit Ratings.

Issuer Rating $\approx$ Senior Unsecured Debt

★ Credit Transition Matrix 基于 Matrix 预估下一年收益率 YTM 改变.

① $\Delta R = \frac{P_1 - P_0}{P_0} = \frac{\Delta P}{P} = \Delta\%P = -\text{Modified Duration} \times \Delta S$

② ΔR 加权(概率)平均.

LM5. Credit Default Swap

CDS

1. 给 Bond 上的保险合同。
2. 标准化 CDS } 投资级 1%
credit spread } 投机级 5% → credit spread standardization 称为 fixed coupon, 如果与 fixed coupon 有差异, 期初一次性多退少补。
3. Long & Short { 若 Bond 违约, $P_B \downarrow$, CDS Seller 赔付, CDS Buyer 获利。→ 本质上 CDS Buyer 看跌, Short 方. Buyer Short
若不违约, $P_B \uparrow$, CDS Seller 不赔付, CDS Seller 获利 → 本质上 CDS Seller 看涨, Long 方. Seller Long
4. Settlement { Physical:
Cash: Seller Payout Amount = Payout Ratio $\times$ Notional Principal 保额
Loss = Par Value - Market Value Seller 方赔付金额。
5. Credit Event Types { Bankruptcy. If only Restructuring, 未破产未违约, 无须赔付。
Failure to pay. (Bond Principal or Interest).

CDS Types

1. Single Name CDS
 - ① 等级 $\neq$ CDS Buyer Bond 的债券违约, 可按 Seller 补偿。
 - ② Cheapest to Deliver (CTD) 原则赔付: 赔付时, 相同级别按 Cheapest Market Value 的 CDS 赔付比例赔付。
2. Index CDS. 成分股 $\rightarrow$ Bond
 - ① Index or Portfolio 内各资产, 平均保额 Notional Amount
 - ② Correlation (ρ) $\uparrow \rightarrow$ risk $\uparrow \rightarrow$ CDS 成本 $\uparrow$.
Buyer Long, Seller Short.

Pricing and Valuation

1. Pricing. ① one period: $CDS\ spread = EL = POD \times LGD = (1 - RR) \times POD$.
② Upfront 期初: 由于 CS 与 Fixed Coupon 差额导致。
 { premium leg 保费 Buyer 付 → For Buyer: Upfront Payment = PV (protection) - PV (premium).
 { protection leg 或有赔付, Buyer 收
 Upfront Premium = (Credit spread - Fixed Coupon 有溢价) $\times$ Duration
 { Seller 退 Buyer 多交的部分 < 0
 { Buyer 补少交的部分 > 0 .
 ③ 报价: $100 - \frac{\text{多退少补 (Upfront Premium)}}{\text{真正}}$
2. Valuation
 $profit\% = \Delta\ spread \times Duration$

Applications

1. Naked CDS: 无 Bond, 买 CDS, 但认为 $P_B \downarrow$, 从而获利。
2. Long/Short Trade 一买一卖套利。
3. Curve Trade
 - Curve Steepen { 短 spread $\downarrow \rightarrow$ 短 risk $\downarrow \rightarrow$ Sell 短 CDS. Long
长 spread $\uparrow \rightarrow$ 长 risk $\uparrow \rightarrow$ Buy 长 CDS. Short
 - Curve Flatten { 短 spread $\uparrow \rightarrow$ 短 risk $\uparrow \rightarrow$ Buy 短 CDS. Short
长 spread $\downarrow \rightarrow$ 长 risk $\downarrow \rightarrow$ Sell 长 CDS. Long
4. Basis Trade Bond credit spread $\neq$ CDS credit spread 时套利。
 { Bond CS $>$ CDS CS $\rightarrow$ Buy Bond (Long) + Buy CDS (Short).
 { Bond CS $<$ CDS CS $\rightarrow$ Sell Bond (Short) + Sell CDS (Long). 无论是否违约, 获利为 CS 之差

